

RWS — Software Logic, Operation Sequence & Fire Control Document

Document Reference: RWS-SW-LOGIC-002

Revision: B

Status: Draft

Classification: Confidential

Parent Document: RWS-SYS-SPEC-001 Rev A

Change from Rev A: Added Section 12A (Mechanical Movement Verification Sensor), updated post-fire sequence, updated signal table, updated flow diagram, added Section 18 (Full System Hardware Architecture).

Table of Contents

1. Purpose & Scope
2. System Context & Key Architectural Assumptions
3. Subsystem Roles & Responsibilities
4. Master / Slave Relationship — Camera vs Pan-Tilt
5. Software State Machine — Full System
6. Phase 1 — System Initialization & Idle
7. Phase 2 — Free Observation Mode (Joystick 1 / Camera Roaming)
8. Phase 3 — Target Lock (Button 1)
9. Phase 4 — Pan-Tilt Slaving to Camera Coordinates
10. Phase 5 — Boresight Achieved / IMU Stabilization Active
11. Phase 6 — Weapon Arming Sequence
12. Phase 7 — Fire Sequence 12A. Mechanical Movement Verification Sensor
13. Phase 8 — Post-Fire, Mechanical Verification & Reset
14. Operator Display Logic
15. Abort & Safety Cutoff Logic
16. Full Sequence Flow Diagram (Text Representation)
17. Interface & Signal Summary
18. Full System Hardware Architecture

1. Purpose & Scope

This document describes in full detail:

- The **software operational logic** of the RWS system
- The **sequence of states** from system power-on to weapon fire
- The **master/slave control relationship** between the electro-optical camera gimbal and the pan-tilt weapon platform
- The **fire control interlock sequence** and all conditions that must be satisfied before solenoid actuation
- The **operator display symbology** during all phases

This document is intended for use by the software development team, system integrator, safety engineer, and verification engineer.

2. System Context & Key Architectural Assumptions

The following architectural facts are foundational to all logic described in this document and must be understood before reading any further.

2.1 Two Separate Platforms — Critical Distinction

The RWS is installed on a **drone platform**. The drone has its own independent flight control system, its own flight camera, and its own operator control — all of which are **entirely separate** from the RWS. The RWS has **no interface with, no dependency on, and no knowledge of** the drone’s flight software, attitude controller, or flight camera. The RWS operates as a **fully autonomous payload** from the drone’s perspective.

2.2 Two Separate Camera Systems

System	Purpose	Controlled by
Drone flight camera	Drone navigation and piloting	Drone operator (separate GCS)
RWS EO/IR camera (on elevation platform)	Target observation and weapon slaving	RWS operator (RWS GCS, Joystick 1)

These two camera systems are **never mixed, never shared, and never co-commanded**.

2.3 Camera Gimbal is a Self-Contained Stabilized Unit

The EO/IR camera assembly is mounted on the **elevation platform** of the RWS pan-tilt system. It has its **own independent gimbal motors and its own internal stabilization system**. This camera gimbal:

- Stabilizes itself against drone vibration and motion using its own internal IMU
- Receives slew commands from the RWS operator via Joystick 1
- Reports its absolute pointing direction (azimuth and elevation angles) to the Jetson FCC at all times
- Acts as the **master** in the master/slave fire control loop

2.4 Pan-Tilt is the Slave

The RWS pan-tilt platform (AZ axis + EL axis) is the **slave** in the fire control loop:

- In normal observation, it remains **stationary at a fixed forward-horizon position**
- It only moves when commanded to slave to the camera’s locked coordinates
- It receives its move commands from the Jetson FCC (not directly from the operator joystick)

2.5 The Jetson FCC is the Brain

The Jetson A (FCC Primary) is the central processing node that:

- Receives camera pointing data continuously
- Receives pan-tilt sensor feedback continuously (AZ heading sensor + EL angle sensor)
- Computes the angular error between camera aim and pan-tilt pointing
- Sends correction commands to the Gimbal Controller AZ servo + EL servo
- Manages the fire control state machine
- Drives all operator display symbology

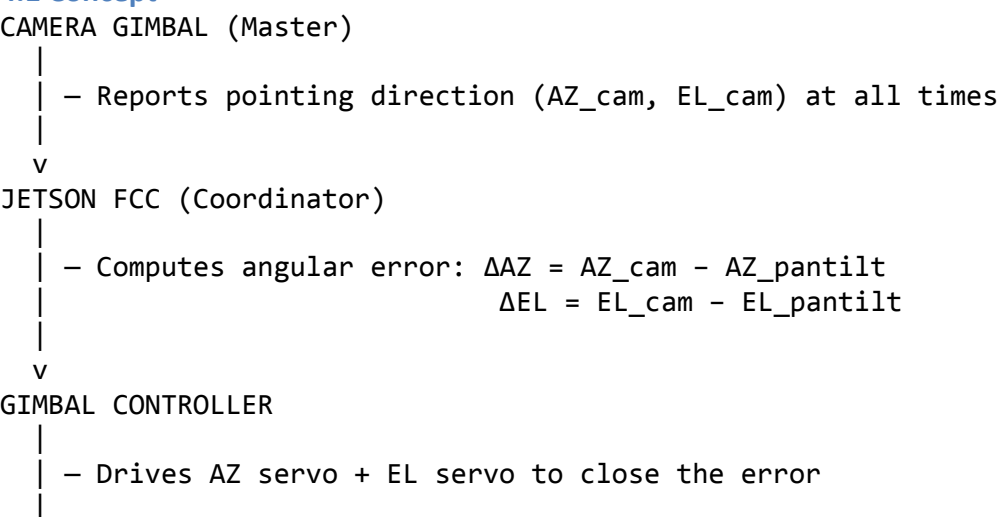
3. Subsystem Roles & Responsibilities

Subsystem	Role in System	Inputs	Outputs
RWS EO/IR Camera Gimbal	Master — observation and target lock	Joystick 1 commands, internal IMU	Camera AZ/EL coordinates to Jetson FCC
Jetson A (FCC Primary)	Central logic, state machine, coordinate processing	Camera coordinates, AZ sensor, EL sensor, operator buttons	Commands to Gimbal Controller, display symbology, ISM authorization

Gimbal Controller	Pan-tilt motor servo controller	Jetson FCC AZ/EL commands, AZ/EL sensor feedback	AZ servo motor command, EL servo motor command
AZ Servo	Azimuth axis actuator	Gimbal Controller CAN command	Azimuth rotation of pan-tilt platform
EL Servo (BLDC, 1:60 worm gear)	Elevation axis actuator	Gimbal Controller CAN command	Elevation rotation of pan-tilt platform
Azimuth Sensor	AZ absolute position feedback	Physical azimuth position	AZ heading angle Jetson FCC
EL Angle Sensor	EL absolute position feedback	Physical elevation position	EL angle Jetson FCC
EL Direction Sensor	EL incremental direction	Physical elevation motion	EL direction Jetson FCC
ISM Safety MCU	Independent safety authority	Jetson FCC authorization signal (discrete), sensors	Solenoid actuation signals
Safety Open Solenoid	Releases weapon safety	ISM command	Mechanical safety open
Safety Close Solenoid	Engages weapon safety	ISM command / failsafe	Mechanical safety close
Safety Pin Solenoid	Master trigger enable	ISM command	Pin actuation
Trigger Solenoid	Weapon fire	ISM command	Trigger actuation
Operator GCS (10")	Human interface	Joystick 1, Button 1, Trigger button	Display output to operator

4. Master / Slave Relationship — Camera vs Pan-Tilt

4.1 Concept



PAN-TILT PLATFORM (Slave)

- Reports its own position via AZ sensor + EL sensor back to Jetson FCC

+-----> (feedback loop closes at Jetson FCC)

4.2 Coordinate Convention

Axis	Zero Reference	Positive Direction	Sensor
Azimuth (AZ)	Forward (nose of drone / front of mount)	Clockwise when viewed from above	Azimuth sensor
Elevation (EL)	Horizontal (0°)	Upward (positive elevation)	EL angle sensor

All angles are expressed in degrees with one decimal place resolution minimum.

4.3 Coordinate Data Flow

Camera Gimbal → [AZ_cam, EL_cam] → Jetson FCC (100 Hz update rate minimum)
AZ Sensor → [AZ_pantilt] → Jetson FCC (100 Hz update rate minimum)
EL Sensor → [EL_pantilt] → Jetson FCC (100 Hz update rate minimum)

Jetson FCC computes:
 $\Delta AZ = AZ_cam - AZ_pantilt$
 $\Delta EL = EL_cam - EL_pantilt$

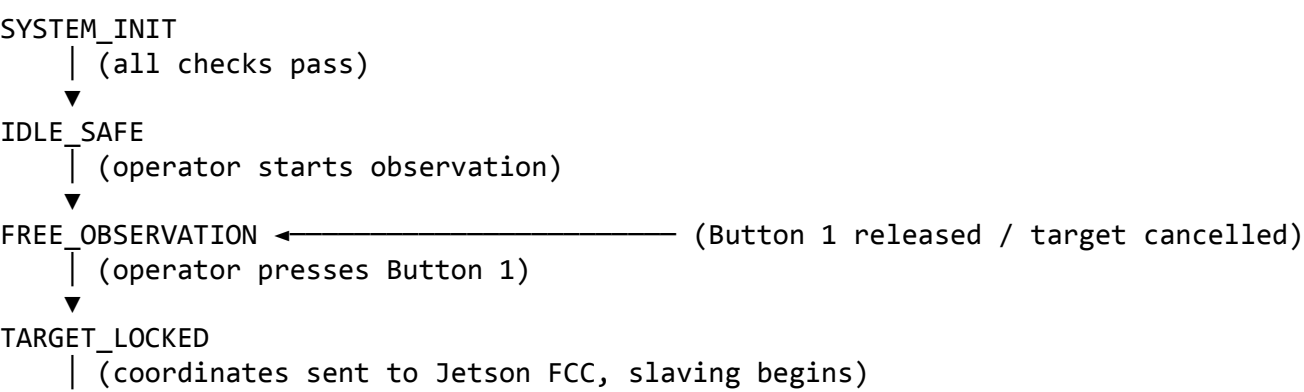
Alignment condition (boresight achieved):
 $|\Delta AZ| \leq 0.5^\circ$ AND $|\Delta EL| \leq 0.5^\circ$ (threshold TBD per weapon accuracy requirement)

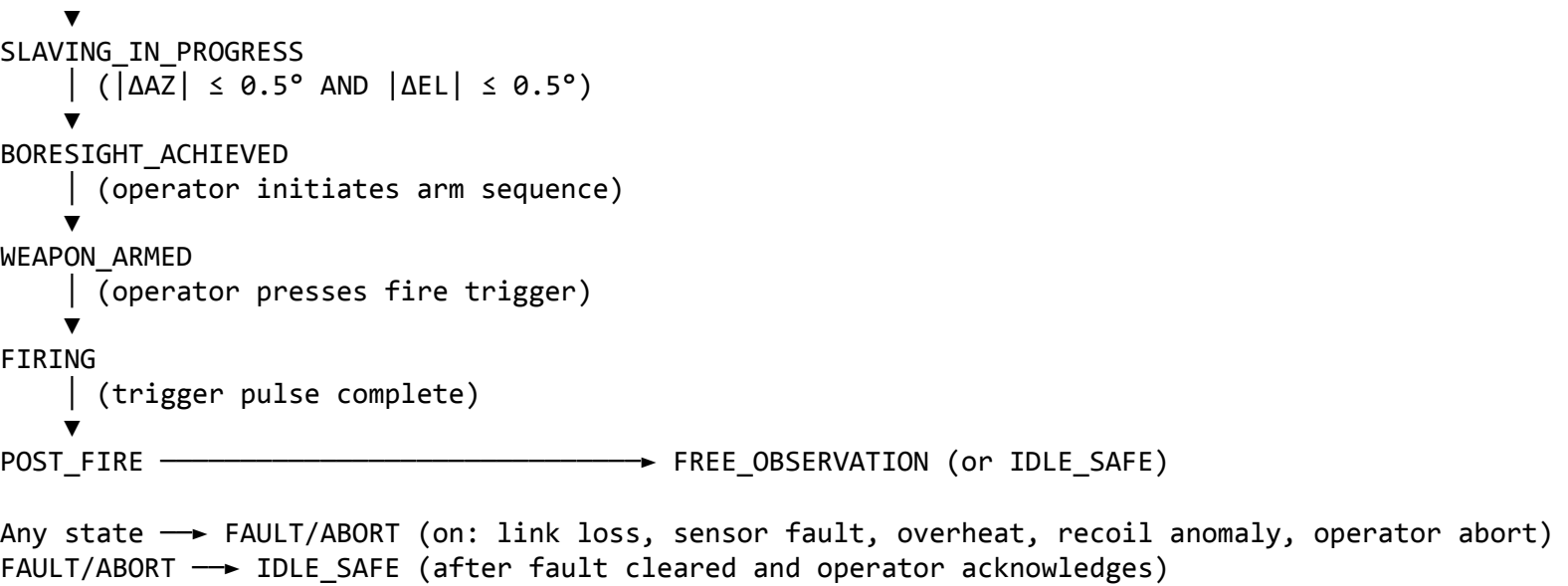
5. Software State Machine — Full System

The Jetson FCC runs a top-level state machine with the following states:

STATE 0 : SYSTEM_INIT
STATE 1 : IDLE_SAFE
STATE 2 : FREE_OBSERVATION
STATE 3 : TARGET_LOCKED
STATE 4 : SLAVING_IN_PROGRESS
STATE 5 : BORESIGHT_ACHIEVED
STATE 6 : WEAPON_ARMED
STATE 7 : FIRING
STATE 8 : POST_FIRE
STATE F : FAULT / ABORT

State transition overview:





6. Phase 1 — System Initialization & Idle

6.1 Power-On Sequence

Upon system power-on, the following initialization sequence executes automatically:

- STEP 1 Power rails stabilize (24V main bus, compute rail, servo rail, safety rail)
- STEP 2 ISM Safety MCU boots – runs self-test independently
 - Confirms all solenoids are de-energized (SAFE state)
 - Confirms weapon temperature within safe range
 - Confirms recoil sensor nominal
- STEP 3 Jetson A boots (≤ 10 seconds to operational)
- STEP 4 Jetson B boots as hot standby
- STEP 5 Gimbal Controller initializes – queries AZ sensor + EL sensor
- STEP 6 EL servo and AZ servo perform homing / reference check
- STEP 7 Camera gimbal initializes its internal stabilization loop
- STEP 8 RF link established with GCS – handshake confirmed
- STEP 9 Optic Master initializes EO camera and IR thermal camera streams
- STEP 10 Jetson A sends "SYSTEM READY" signal to GCS

6.2 Pan-Tilt Default Position

Upon completion of initialization, the pan-tilt platform **moves to and locks at the fixed forward-horizon position**:

AZ_default = 0.0° (forward)
EL_default = 0.0° (horizon)

The pan-tilt holds this position rigidly using the worm gear's self-locking property. The EL and AZ servo drives are powered but commanded to hold position. The platform does **not respond to Joystick 1** — the joystick controls only the camera gimbal at this stage.

6.3 Entry Conditions to IDLE_SAFE

- All solenoids confirmed de-energized
- ISM in SAFE state
- Pan-tilt at forward-horizon default
- Camera streams active on GCS display
- RF link confirmed with latency ≤ 50 ms

7. Phase 2 — Free Observation Mode (Joystick 1 / Camera Roaming)

7.1 Description

In FREE_OBSERVATION state, the operator uses **Joystick 1** to freely slew the EO/IR camera gimbal in all directions — azimuth and elevation — to observe the area of interest. The camera moves independently on its own stabilized gimbal.

7.2 Pan-Tilt Behavior

- The pan-tilt platform **remains stationary at the forward-horizon default position (AZ=0°, EL=0°)**
- The pan-tilt does not track the camera in this phase
- The pan-tilt motors are powered and holding position via servo lock

7.3 Camera Behavior

- Camera gimbal receives Joystick 1 slew commands
- Camera gimbal stabilizes its own pointing against drone vibration and motion
- Camera reports its current pointing angles (AZ_cam, EL_cam) to Jetson FCC at ≥ 100 Hz
- Operator views live EO or IR video feed on GCS 10" display
- The operator can switch between EO (day) and IR (thermal) feeds at any time

7.4 Jetson FCC Behavior

- Continuously receives and logs (AZ_cam, EL_cam) from camera
- Continuously receives and logs (AZ_pantilt, EL_pantilt) from AZ and EL sensors
- Does **not** issue any slaving commands in this state
- Monitors all system health parameters
- Monitors RF link quality

7.5 Operator Display — FREE_OBSERVATION

```
LIVE VIDEO FEED (EO or IR)

Crosshair at camera center (white)

AZ_cam: 034.5°    EL_cam: -12.3°
PAN-TILT: HOLD    FORWARD/HORIZON

STATE: FREE OBSERVATION
WEAPON: SAFE
```

8. Phase 3 — Target Lock (Button 1)

8.1 Trigger Condition

The operator identifies a target of interest via the camera feed and **presses Button 1**.

8.2 Camera Behavior on Button 1

Upon Button 1 press, the camera gimbal executes the following immediately:

- STEP 1 Camera freezes its current pointing direction
AZ_lock = AZ_cam at moment of button press
EL_lock = EL_cam at moment of button press
- STEP 2 Camera activates internal target lock / position hold mode
→ Camera stabilizes on (AZ_lock, EL_lock) – does NOT follow any further joystick input
→ Camera internal stabilization loop now holds this absolute pointing direction
- STEP 3 Camera reports confirmed lock coordinates to Jetson FCC:
MESSAGE: TARGET_LOCK_CONFIRMED { AZ_lock, EL_lock, timestamp }

8.3 Jetson FCC Behavior on TARGET_LOCK_CONFIRMED

```
STEP 1 Jetson FCC receives and stores (AZ_lock, EL_lock)
STEP 2 Jetson FCC transitions state machine to STATE 3: TARGET_LOCKED
STEP 3 Jetson FCC sends locked coordinates to Gimbal Controller as the slaving target
STEP 4 Jetson FCC sends display update to GCS:
    → Draw a WHITE SQUARE around the locked target position on the video feed
    → Display locked coordinates numerically on the HUD
```

8.4 Operator Display — TARGET_LOCKED

LIVE VIDEO FEED (EO or IR)

TARGET

← White square on target

TARGET LOCKED

AZ_lock: 034.5° EL_lock: -12.3°

PAN-TILT: SLAVING INITIATED

WEAPON: SAFE

9. Phase 4 — Pan-Tilt Slaving to Camera Coordinates

9.1 Description

Immediately after TARGET_LOCK_CONFIRMED, the Jetson FCC commands the Gimbal Controller to move the pan-tilt platform from its current position (forward-horizon) toward the locked camera coordinates (AZ_lock, EL_lock). The pan-tilt now acts as the slave, closing the angular error gap.

During this slaving phase the IMU is **not yet active** as a stabilization source — the pan-tilt is in gross motion and IMU stabilization only activates once boresight is achieved (see Phase 5).

9.2 Slaving Control Logic (Jetson FCC)

LOOP (running at ≥ 100 Hz):

```
READ AZ_pantilt from azimuth sensor
READ EL_pantilt from EL angle sensor
```

COMPUTE:

```
ΔAZ = AZ_lock - AZ_pantilt
ΔEL = EL_lock - EL_pantilt
```

IF $|\Delta AZ| > 0.5^\circ$ OR $|\Delta EL| > 0.5^\circ$:

SEND to Gimbal Controller:

```
AZ_command = AZ_lock    (absolute position target)
EL_command = EL_lock    (absolute position target)
```

IMU_STABILIZATION = OFF

STATE = SLAVING_IN_PROGRESS

IF $|\Delta AZ| \leq 0.5^\circ$ AND $|\Delta EL| \leq 0.5^\circ$:

SEND to Gimbal Controller: HOLD_POSITION

STATE = BORESIGHT_ACHIEVED (go to Phase 5)

END LOOP

9.3 Gimbal Controller Behavior

The Gimbal Controller executes the position servo loop:

FOR AZ axis:

Error_AZ = AZ_command - AZ_sensor_feedback
AZ_motor_velocity = Kp_AZ × Error_AZ + Kd_AZ × d(Error_AZ)/dt
Send CAN velocity command to AZ servo drive

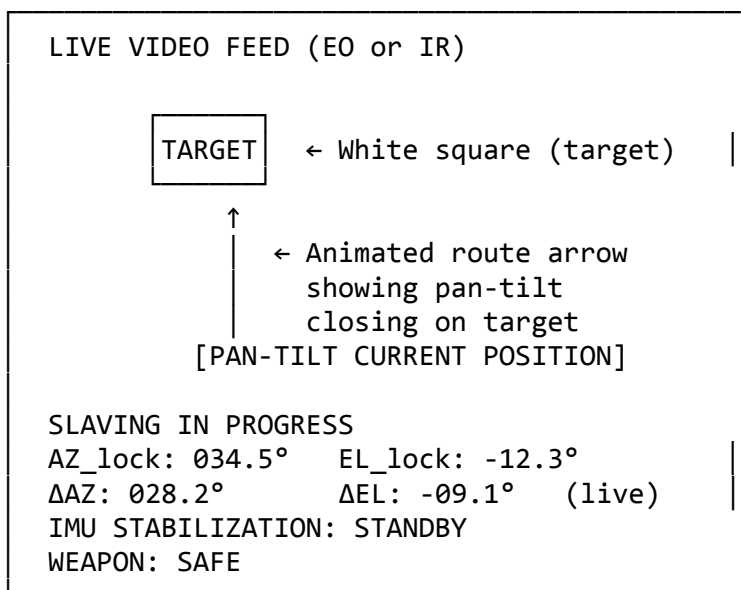
FOR EL axis:
Error_EL = EL_command - EL_sensor_feedback
EL_motor_velocity = Kp_EL × Error_EL + Kd_EL × d(Error_EL)/dt
Send CAN velocity command to EL servo drive (BLDC, 1:60 worm gear)

9.4 Pan-Tilt Motion Constraints

Parameter	Value
Maximum AZ slew speed	≥ 22.5 °/s (MIL-STD-1472H)
Maximum EL slew speed	Full arc ≤ 3 seconds (MIL-STD-1472H)
Motion profile	Trapezoidal velocity profile (controlled acceleration and deceleration)
Overshoot	≤ 0.5° on both axes
Approach behavior	Decelerate as

9.5 Operator Display — SLAVING_IN_PROGRESS

During pan-tilt slaving, the GCS display shows the operator a **live routing indicator**:



Display elements during SLAVING_IN_PROGRESS: - White square remains on locked target position - An animated arrow or progress indicator moves from the pan-tilt's current position toward the target square - Live ΔAZ and ΔEL values displayed numerically and decrease in real time - State label: "SLAVING IN PROGRESS" - IMU status: "STANDBY" (not yet active) - Weapon status remains: SAFE

10. Phase 5 — Boresight Achieved / IMU Stabilization Active

10.1 Trigger Condition

$|\Delta AZ| \leq 0.5^\circ$ AND $|\Delta EL| \leq 0.5^\circ$

Both conditions must be true **simultaneously and sustained for ≥ 500 ms** before boresight is declared confirmed. This prevents false confirmation due to mechanical overshoot or sensor noise.

10.2 IMU Stabilization Loop — Activation

This is the most critical change from Phase 4. The moment boresight is confirmed, the Jetson FCC **activates the IMU stabilization loop** for the pan-tilt AZ and EL servos. From this point forward, the IMU continuously measures the drone's angular rates and body motion, and the Jetson FCC uses these measurements to issue real-time correction commands to the AZ and EL servos, keeping the weapon pointed at (AZ_lock, EL_lock) in **world-fixed space** regardless of drone movement.

The IMU stabilization loop operates as follows:

IMU provides at ≥ 1 kHz:

```
 $\omega_x$  = roll rate      (rad/s)
 $\omega_y$  = pitch rate   (rad/s)
 $\omega_z$  = yaw rate     (rad/s)
```

Jetson FCC IMU Stabilization Loop (runs at ≥ 500 Hz):

```
READ IMU angular rates: [ $\omega_x$ ,  $\omega_y$ ,  $\omega_z$ ]
```

```
TRANSFORM drone angular rates to pan-tilt axis corrections:
```

```
 $\delta AZ_{imu} = -\omega_z \times dt$           (yaw of drone disturbs AZ directly)
 $\delta EL_{imu} = -(\omega_y \times \cos(AZ_{lock}) - \omega_x \times \sin(AZ_{lock})) \times dt$ 
                    (pitch and roll of drone disturb EL depending on current AZ)
```

```
COMPUTE corrected servo target:
```

```
 $AZ_{corrected} = AZ_{lock} + \Sigma(\delta AZ_{imu})$     (integrated correction)
 $EL_{corrected} = EL_{lock} + \Sigma(\delta EL_{imu})$     (integrated correction)
```

```
SEND to Gimbal Controller:
```

```
 $AZ_{command} = AZ_{corrected}$ 
 $EL_{command} = EL_{corrected}$ 
```

```
VERIFY via sensors:
```

```
READ  $AZ_{pantilt}$ ,  $EL_{pantilt}$ 
Residual_AZ =  $AZ_{corrected} - AZ_{pantilt}$ 
Residual_EL =  $EL_{corrected} - EL_{pantilt}$ 
→ Servo loop closes the residual at  $\geq 100$  Hz
```

```
IF  $|Residual\_AZ| > 1.5^\circ$  OR  $|Residual\_EL| > 1.5^\circ$ :
```

```
→ Drone disturbance exceeds servo correction capability
→ Jetson FCC sends STABILIZATION_WARNING to GCS
→ IF sustained > 2 seconds: exit BORESIGHT_ACHIEVED → re-slave
```

```
END LOOP
```

10.3 IMU Stabilization Loop — Key Properties

Property	Value
IMU data rate	≥ 1 kHz
Stabilization correction rate	≥ 500 Hz
Servo position correction rate	≥ 100 Hz
Axes corrected	AZ (yaw) + EL (pitch/roll coupled)
Maximum correctable drone disturbance	$\pm 5^\circ/s$ angular rate (servo bandwidth limited)
Stabilization residual error	$\leq 0.3^\circ$ RMS under normal drone hover disturbance
IMU drift correction	Low-pass filter on integrated IMU bias to prevent drift

10.4 IMU Stabilization Loop — Deactivation Condition

The IMU stabilization loop is **active only when TARGET_LOCK is engaged**. It is immediately deactivated under any of the following conditions:

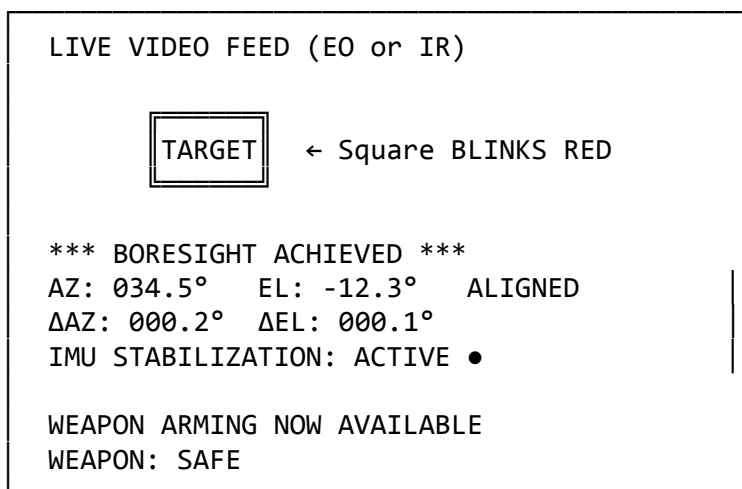
Condition	IMU Loop Action
Operator releases TARGET LOCK (Button 1 pressed again)	STOP — IMU loop opens, pan-tilt returns to forward-horizon
System enters FAULT / ABORT state	STOP — pan-tilt holds last position passively
System returns to FREE_OBSERVATION state	STOP — IMU loop closed
Drone angular rate exceeds $\pm 15^\circ/\text{s}$ (structural limit)	STOP — stabilization warning issued, re-slave required

When the IMU loop stops, the pan-tilt **does not move abruptly** — the Gimbal Controller receives a HOLD_LAST_POSITION command and the worm gear passive self-lock maintains position until the next command.

10.5 Jetson FCC Actions on Boresight Achieved

- STEP 1 Jetson FCC transitions to STATE 5: BORESIGHT_ACHIEVED
- STEP 2 Jetson FCC activates IMU stabilization loop
 - IMU corrections begin feeding into AZ and EL servo commands
 - Worm gear provides passive mechanical hold between corrections
- STEP 3 Jetson FCC sends BORESIGHT_ACHIEVED event to GCS display
 - "IMU STABILIZATION: ACTIVE" shown on HUD in green
- STEP 4 Jetson FCC sends BORESIGHT_ACHIEVED to ISM Safety MCU (discrete signal)
 - ISM acknowledges but does NOT arm yet
 - ISM awaits further operator arm command

10.6 Operator Display — BORESIGHT_ACHIEVED + IMU ACTIVE



Key display changes: - Target square changes from white static to **blinking red** — boresight confirmed - New HUD element: **"IMU STABILIZATION: ACTIVE"** in green with a pulsing green dot — this tells the operator the pan-tilt is actively compensating for drone motion - The operator can observe the pan-tilt micro-corrections happening in real time via the ΔAZ/ΔEL values remaining near zero despite drone movement

11. Phase 6 — Weapon Arming Sequence

11.1 Entry Condition

Weapon arming is **only permitted** when **ALL** of the following conditions are simultaneously true:

Condition	Source	Required State
STATE =	Jetson FCC state	True

BORESIGHT_ACHIEVE	machine	
D		
	ΔAZ	$\leq 0.5^\circ$
	ΔEL	$\leq 0.5^\circ$
Weapon temperature $\leq$ T_max	ISM Safety MCU	True
Recoil sensor nominal	ISM Safety MCU	True
RF link active, latency $\leq$ 50 ms	Jetson FCC	True
ISM Safety MCU in SAFE state	ISM	True

If **any** of the above conditions is not met, the arm command from the operator is **rejected** and the display shows a fault reason.

11.2 Arm Command Sequence

The operator initiates arming by pressing the **Arm Button** (dedicated hardware button, not touchscreen):

- STEP 1 Operator presses Arm Button
- STEP 2 Jetson FCC checks all entry conditions (table above)
- STEP 3 IF all conditions met:
 - Jetson FCC sends ARMING_AUTHORIZATION (discrete signal) to ISM Safety MCUELSE:
 - Jetson FCC sends ARMING_REJECTED + fault reason to GCS display
 - Remain in BORESIGHT_ACHIEVED state
- STEP 4 ISM Safety MCU receives ARMING_AUTHORIZATION
ISM independently verifies:
 - Weapon temperature check
 - Recoil sensor check
 - All discrete safety inputs nominal
- STEP 5 IF ISM verification passes:
 - ISM energizes SAFETY OPEN SOLENOID → opens weapon mechanical safety
 - ISM energizes SAFETY PIN SOLENOID → arms master trigger
 - ISM transitions to ARMED state
 - ISM sends ARMED_CONFIRMED (discrete signal) to Jetson FCC
- STEP 6 Jetson FCC receives ARMED_CONFIRMED
Jetson FCC transitions to STATE 6: WEAPON_ARMED
Jetson FCC sends ARMED display update to GCS

11.3 Operator Display — WEAPON_ARMED

LIVE VIDEO FEED (EO or IR)

TARGET

← Square BLINKS RED

*** WEAPON ARMED ***

AZ: 034.5° EL: -12.3° ALIGNED

SAFETY: OPEN

TRIGGER: ARMED

READY TO FIRE

12. Phase 7 — Fire Sequence

12.1 Entry Condition

Fire is **only permitted** when **ALL** of the following are simultaneously true:

Condition	Source	Required State
STATE = WEAPON_ARMED	Jetson FCC	True
ISM in ARMED state	ISM Safety MCU	True
	ΔAZ	$\leq 0.5^\circ$ (still aligned)
	ΔEL	$\leq 0.5^\circ$ (still aligned)
RF link active	Jetson FCC	True
Weapon temperature $\leq T_{max}$	ISM Safety MCU	True

If alignment drifts beyond 0.5° after arming (e.g., due to drone disturbance), the system **automatically exits WEAPON_ARMED** back to SLAVING_IN_PROGRESS and re-slaves until alignment is recovered before the operator can fire again.

12.2 Fire Command Sequence

- STEP 1 Operator presses Trigger Button (dedicated physical hardware button)
- STEP 2 Jetson FCC checks all fire entry conditions
- STEP 3 IF all conditions met:
 Jetson FCC sends FIRE_AUTHORIZATION (discrete signal) to ISM Safety MCU
ELSE:
 Fire command rejected – display reason to operator
- STEP 4 ISM Safety MCU receives FIRE_AUTHORIZATION
ISM independently verifies all armed conditions one final time (< 5 ms check)
- STEP 5 IF ISM verification passes:
 ISM pulses TRIGGER SOLENOID for the weapon-defined pulse duration
 → Trigger pin actuated → weapon fires
- STEP 6 ISM monitors RECOIL MOVEMENT SENSOR:
 IF recoil pulse detected within expected window:
 → Fire confirmed (mechanical verification)
 → ISM logs FIRE_CONFIRMED event
 ELSE:
 → Misfire detected → ISM sends MISFIRE_ALERT to Jetson FCC
- STEP 7 Jetson FCC transitions to STATE 7: FIRING → STATE 8: POST_FIRE

12.3 Trigger Solenoid Pulse Parameters

Parameter	Value
Pulse voltage	24 VDC
Pulse duration	Defined by weapon platform (typical: 50–200 ms)
Maximum duty cycle	Single pulse per fire command
ISM verification window	Within 500 ms post-trigger

12A. Mechanical Movement Verification Sensor

12A.1 Purpose & Concept

After every fire sequence, the weapon mechanism must complete a defined mechanical cycle — bolt carrier travel, ejection, and reload. A **dedicated short-travel displacement sensor** monitors a **20 mm mechanical movement** on the weapon action to verify that this full cycle has been completed correctly. This sensor is the gate that authorizes or blocks the next fire sequence.

This is a critical safety and reliability function. If the mechanical cycle did not complete — due to a misfire, jam, feed failure, or mechanical fault — the system **blocks all further fire commands** and alerts the operator immediately with a full fault description.

12A.2 Sensor Specification

Parameter	Requirement
Sensor type	Linear displacement sensor (inductive proximity, Hall-effect magnetic, or LVDT)
Measurement range	0 – 25 mm (covers full 20 mm travel with margin)
Required travel confirmation	20 mm ± 1 mm
Response time	≤ 5 ms detection latency
Output signal	Discrete digital pulse (HIGH = travel complete) OR analog 0–5V
Interface	Discrete signal ISM Safety MCU
Operating environment	IP54 minimum, vibration-rated
Mounting	Fixed to weapon frame, sensing element aligned to moving bolt/carrier component
Fail state	No signal = fault (active confirmation required, absence = block)

12A.3 Detection Logic — ISM Safety MCU

The ISM Safety MCU is the sole receiver and interpreter of the mechanical movement sensor signal. The logic is as follows:

AFTER TRIGGER SOLENOID PULSE:

START verification timer T_mech (timeout = 300 ms)

MONITOR mechanical movement sensor:

CASE 1 – MOVEMENT CONFIRMED:

Sensor detects displacement ≥ 20 mm within T_mech window

- ISM sets flag: MECH_CYCLE_OK = TRUE
- ISM sends MECH_CYCLE_CONFIRMED to Jetson FCC
- GCS display: "CYCLE OK – READY FOR NEXT FIRE" in green
- Next fire sequence is AUTHORIZED

CASE 2 – MOVEMENT NOT DETECTED (Timeout):

Sensor does NOT detect 20 mm displacement within T_mech = 300 ms

- ISM sets flag: MECH_CYCLE_OK = FALSE
- ISM sets fault: MECH_FAULT = JAM / INCOMPLETE_CYCLE
- ISM sends MECH_CYCLE_FAULT to Jetson FCC
- ISM blocks all further ARMING_AUTHORIZATION signals
- GCS display: FAULT BANNER – "WEAPON FAULT: MECHANICAL CYCLE INCOMPLETE"
- Next fire sequence is BLOCKED until fault is cleared

CASE 3 – PARTIAL MOVEMENT (sensor detects < 20 mm):
Sensor detects movement but < 20 mm threshold
→ Treated as CASE 2 (incomplete cycle)
→ ISM sets fault: MECH_FAULT = PARTIAL_CYCLE
→ Same blocking and alert behavior as CASE 2

12A.4 Fault Classification & Operator Messages

Fault Code	Sensor Reading	Displayed Message	Cause
MECH-OK	≥ 20 mm within 300 ms	“CYCLE OK — READY FOR NEXT FIRE” (green)	Normal operation
MECH-JAM-01	0 mm — no movement detected	“WEAPON FAULT: NO MECHANICAL MOVEMENT — POSSIBLE JAM” (red flashing)	Complete jam or feed failure
MECH-JAM-02	< 20 mm — partial movement	“WEAPON FAULT: INCOMPLETE CYCLE — PARTIAL MOVEMENT DETECTED” (red flashing)	Partial jam or weak actuation
MECH-TIMEOUT	No signal within 300 ms	“WEAPON FAULT: MECHANICAL VERIFICATION TIMEOUT” (red flashing)	Sensor fault or disconnection

12A.5 Fault Clearance Procedure

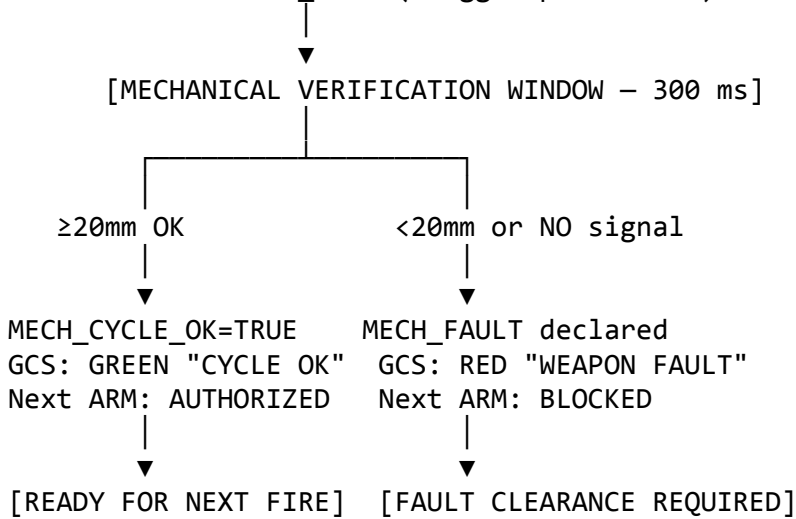
When a MECH_FAULT is declared, the following conditions must ALL be met before the next fire sequence can be authorized:

STEP 1 Operator physically clears the weapon jam / confirms weapon state
STEP 2 Operator presses FAULT ACKNOWLEDGE on GCS (dedicated button or touchscreen)
STEP 3 ISM Safety MCU performs sensor re-check:
→ Verifies mechanical movement sensor is reading zero (weapon in battery / closed position)
→ Verifies weapon temperature within limits
→ Verifies recoil sensor nominal
STEP 4 IF all re-checks pass:
ISM clears MECH_FAULT flag
ISM sends MECH_FAULT_CLEARED to Jetson FCC
System returns to BORESIGHT_ACHIEVED state (camera still locked)
Operator may re-arm and fire
ELSE:
ISM remains in FAULT state
GCS continues to display fault banner
Operator must physically inspect weapon before next attempt

12A.6 Integration Into Fire Sequence — Entry Gate

The mechanical movement sensor status creates a new **mandatory gate** in the fire sequence:

PREVIOUS STATE: POST_FIRE (trigger pulse sent)



12A.7 System Initialization Self-Test

During SYSTEM_INIT (Phase 1), the ISM Safety MCU performs a self-test on the mechanical movement sensor:

- STEP 1 ISM reads mechanical movement sensor at power-on
- STEP 2 IF sensor reads 0 mm (weapon in closed/battery position): SENSOR_OK → continue boot
- STEP 3 IF sensor reads non-zero at rest: SENSOR_FAULT → boot halted, alert displayed
- STEP 4 Sensor is confirmed present and responding before IDLE_SAFE state is entered

13. Phase 8 — Post-Fire, Mechanical Verification & Reset

13.1 Immediate Post-Fire Actions (ISM Safety MCU)

- STEP 1 ISM de-energizes TRIGGER SOLENOID immediately after pulse
- STEP 2 ISM de-energizes SAFETY PIN SOLENOID → master trigger disarmed
- STEP 3 ISM energizes SAFETY CLOSE SOLENOID → closes weapon mechanical safety
- STEP 4 ISM transitions to SAFE state
- STEP 5 ISM STARTS mechanical verification window ($T_{\text{mech}} = 300 \text{ ms}$)
→ Monitors mechanical movement sensor for $\geq 20 \text{ mm}$ displacement
- STEP 6 ISM evaluates result (see Section 12A.3):
 - IF MECH_CYCLE_OK = TRUE: MECH_CYCLE_CONFIRMED sent to Jetson FCC
 - IF MECH_CYCLE_OK = FALSE: MECH_FAULT declared, sent to Jetson FCC
- STEP 7 ISM sends SAFE_CONFIRMED (discrete) to Jetson FCC

13.2 Post-Fire Actions (Jetson FCC)

- STEP 1 Jetson FCC receives SAFE_CONFIRMED from ISM
- STEP 2 Jetson FCC logs fire event with timestamp, AZ, EL, target coordinates
- STEP 3 Jetson FCC reads WEAPON TEMPERATURE – if above T_{warning} : flag for operator
- STEP 4 Jetson FCC transitions to STATE 8: POST_FIRE
- STEP 5 IMU STABILIZATION LOOP REMAINS ACTIVE during post-fire observation window
→ Camera gimbal remains locked on target (for hit/miss confirmation per MIL-STD-1472H §5.13.1.5.1.4)
→ Pan-tilt continues to be held on target coordinate by IMU corrections

- STEP 6 Jetson FCC receives MECH_CYCLE result from ISM:

CASE A – MECH_CYCLE_CONFIRMED (cycle OK):

- GCS displays "CYCLE OK – READY FOR NEXT FIRE" in green
- System confirms: next ARM sequence is authorized
- Operator may re-arm without clearance procedure
- State: BORESIGHT_ACHIEVED maintained (camera still locked)

CASE B – MECH_FAULT received (cycle incomplete):

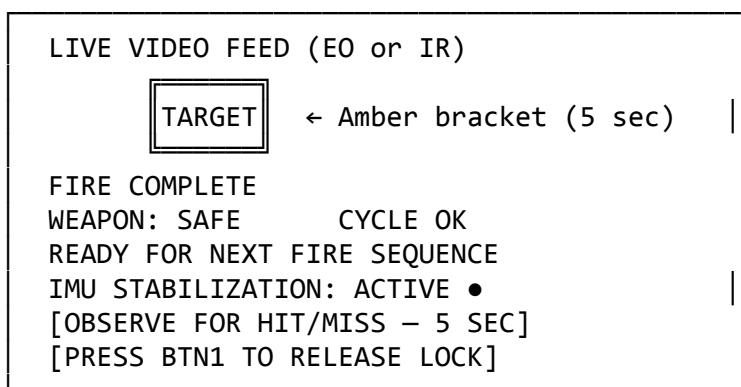
- GCS displays RED FAULT BANNER with fault code and message

- System BLOCKS next ARMING_AUTHORIZATION
- IMU stabilization remains active (camera remains on target)
- Operator must acknowledge fault and follow clearance procedure (Section 12A.5)

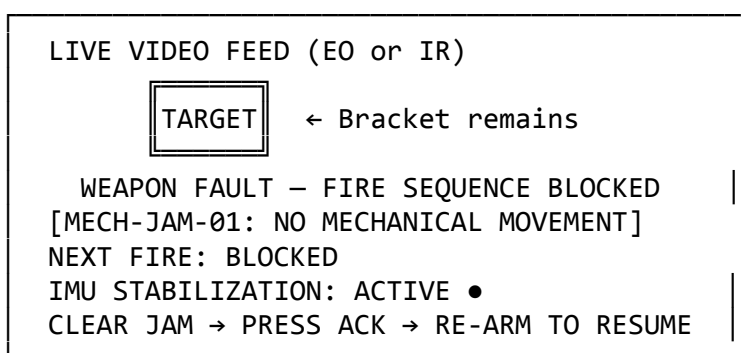
STEP 7 After operator acknowledges TARGET LOCK release (Button 1 again) or 5-second timeout:
*** IMU STABILIZATION LOOP DEACTIVATES ***
Gimbal Controller receives HOLD_LAST_POSITION
Worm gear passive self-lock holds pan-tilt at last position
Jetson FCC transitions to STATE 2: FREE_OBSERVATION
→ Camera lock is released → operator regains Joystick 1 control
→ Pan-tilt may be commanded back to forward-horizon or remain at current position

13.3 Operator Display — POST_FIRE

Case A — Mechanical cycle confirmed (normal):



Case B — Mechanical fault (cycle incomplete):



14. Operator Display Logic

14.1 HUD Symbol Summary

Symbol	Color	Meaning	Phase
Crosshair (center)	White	Camera pointing direction	All phases
Target square	White (static)	Target locked by camera	Phase 3
Route arrow	White animated	Pan-tilt slaving in progress	Phase 4
Δ AZ / Δ EL readout	White	Live angular error values	Phase 4
“IMU STABILIZATION:	Grey	IMU not yet active	Phase 4

STANDBY"			
Target square	Red (blinking)	Boresight achieved — aligned	Phase 5 onwards
"IMU STABILIZATION: ACTIVE "	Green pulsing	IMU actively correcting AZ/EL servos	Phase 5, 6, 7, 8
"WEAPON ARMED" banner	Red	System armed, ready to fire	Phase 6
"CYCLE OK — READY FOR NEXT FIRE"	Green	Mechanical cycle confirmed	Phase 8 normal
"WEAPON FAULT: [code]"	Red flashing	Mechanical cycle incomplete — next fire blocked	Phase 8 fault
"FIRE COMPLETE" banner	Amber	Post-fire confirmation	Phase 8
"IMU STABILIZATION: OFF"	Grey	Target lock released, IMU loop closed	FREE_OBSERVATION
Fault banner	Flashing orange	Any fault or abort condition	Phase F

14.2 Pan-Tilt Route Display Logic

During SLAVING_IN_PROGRESS:

- The GCS overlays the current pan-tilt position as a secondary crosshair or dot on the compass/angular display
- An animated line or arrow connects the pan-tilt current position to the target locked square
- As ΔAZ and ΔEL decrease, the animated indicator visually closes the gap toward the target
- Numerical ΔAZ and ΔEL update in real time (minimum 10 Hz refresh)

14.3 Alignment Confirmation Indication

When boresight is achieved:

- The target square changes to **blinking red at 2 Hz**
- The ΔAZ and ΔEL readouts change to green and show values $\leq$ threshold
- An audio tone (if implemented) sounds once
- The "WEAPON ARMING NOW AVAILABLE" label appears

15. Abort & Safety Cutoff Logic

15.1 Abort Triggers

The system **immediately transitions to FAULT/ABORT state** on any of the following events, regardless of current state:

Event	Source	System Response
RF link loss > 2 seconds	Jetson FCC RF monitor	Weapon SAFE, servos hold position
Jetson A watchdog timeout	Hardware watchdog	Failover to Jetson B; weapon SAFE

Weapon temperature > T_cutoff	ISM Safety MCU	ISM forces SAFE, fire rejected
Recoil sensor anomaly (unexpected fire)	ISM Safety MCU	ISM forces SAFE, MISFIRE alert
Mechanical movement sensor fault (no signal at boot)	ISM Safety MCU	Boot halted, sensor fault alert
MECH_FAULT — incomplete cycle post-fire	ISM Safety MCU	Next fire BLOCKED, fault banner displayed
AZ/EL sensor fault	Jetson FCC	Servos halt, weapon SAFE
ISM heartbeat loss	Jetson FCC	Weapon SAFE, operator alert
Operator presses ABORT button	GCS	Immediate SAFE state
Alignment lost after BORESIGHT ($ \Delta > 0.5^\circ$)	Jetson FCC	Exit ARMED, re-slave required

15.2 Abort Sequence

```

STEP 1  Abort trigger detected
STEP 2  Jetson FCC sends ABORT command to ISM (discrete signal)
STEP 3  ISM immediately:
        → De-energizes all solenoids
        → Energizes SAFETY CLOSE solenoid
        → Transitions to SAFE state
STEP 4  Gimbal Controller: AZ and EL servos commanded to hold position (no runaway)
STEP 5  GCS display: Flashing orange FAULT banner with fault reason text
STEP 6  Camera: Remains locked on last position (if camera lock was active)
STEP 7  System remains in FAULT state until:
        → Fault condition cleared
        → Operator manually acknowledges fault on GCS
        → System transitions to IDLE_SAFE

```

15.3 Non-Overridable Safety Rules

The following rules are **hardcoded in the ISM Safety MCU and cannot be overridden by software**:

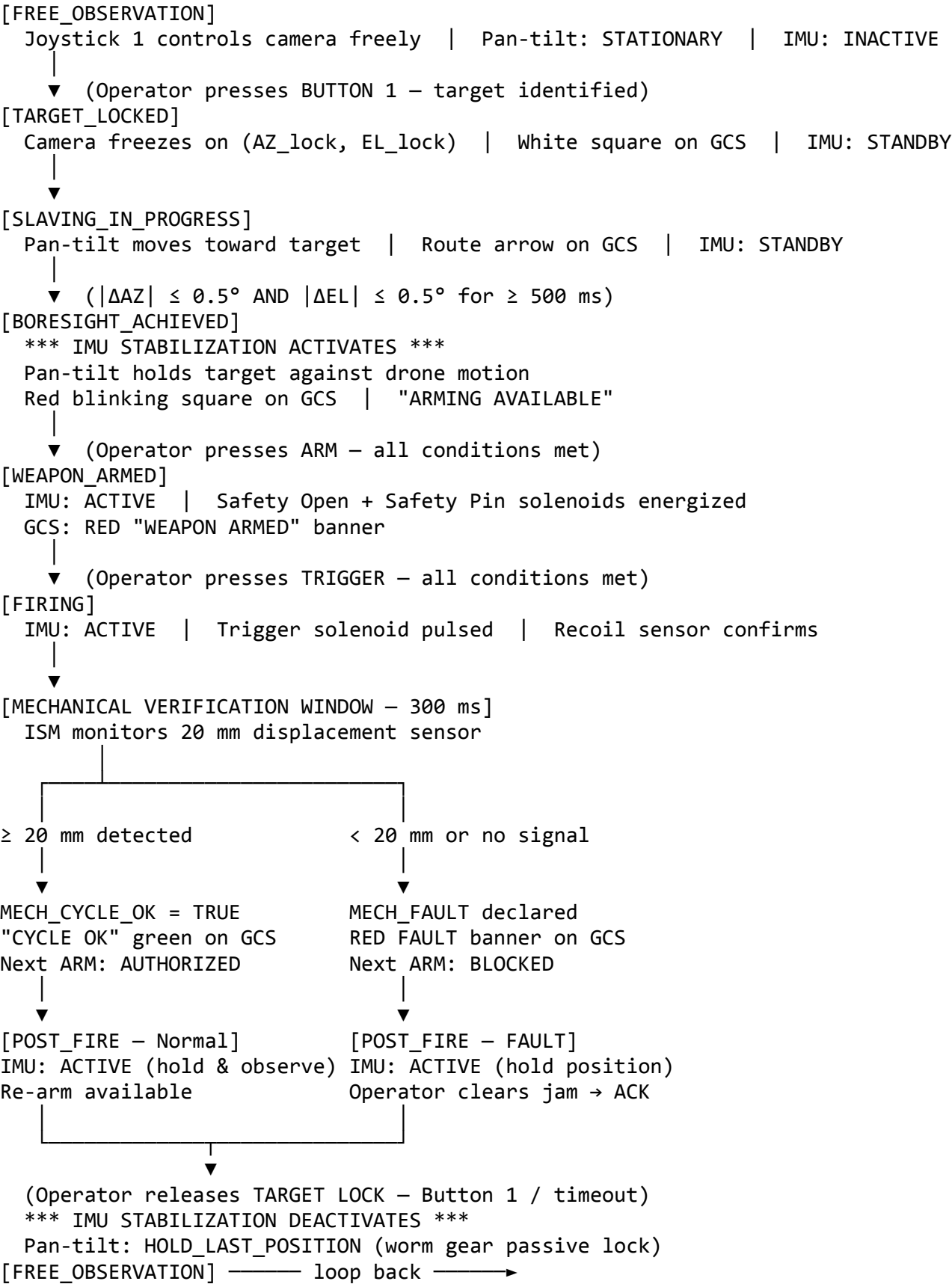
1. The trigger solenoid can **never** be energized unless the ISM is in ARMED state
2. The ARMED state can **never** be entered unless the BORESIGHT_ACHIEVED signal is present
3. Any loss of the ISM's own heartbeat causes all solenoids to de-energize (fail-safe design)
4. De-energized solenoid = SAFE mechanical state (spring-return to safe)

16. Full Sequence Flow Diagram (Text Representation)

```

POWER ON
  |
  ▼
[SYSTEM INIT]
All boot checks pass – mechanical movement sensor self-test included
Pan-tilt moves to AZ=0° / EL=0° (forward-horizon)
  |
  ▼
[IDLE_SAFE]
Pan-tilt: HOLD at 0°/0° | Weapon: SAFE | IMU: INACTIVE
  |
  ▼ (Operator activates observation)

```



Two governing rules — always apply: 1. TARGET_LOCK engaged IMU stabilization ACTIVE. TARGET_LOCK released INACTIVE. 2. MECH_CYCLE_OK = TRUE next fire AUTHORIZED. MECH_CYCLE_OK = FALSE next fire BLOCKED.

17. Interface & Signal Summary

Signal Name	Direction	Type	Description
AZ_cam, EL_cam	Camera	Jetson Ethernet (100	Camera current

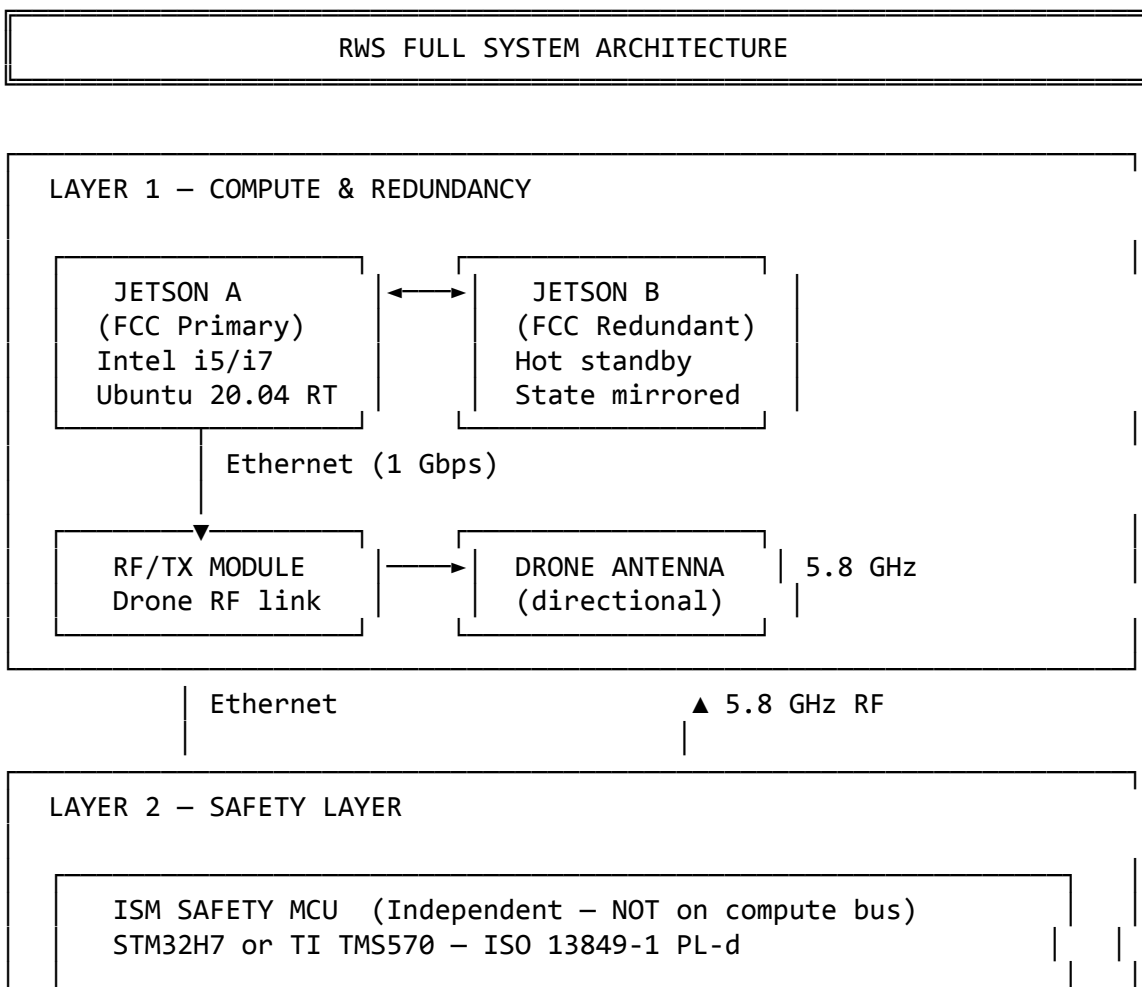
	FCC		Hz)	pointing angles
TARGET_LOCK_CONFIRMED	Camera FCC	Jetson	Ethernet	Lock coordinates confirmed
IMU angular rates	IMU FCC	Jetson	SPI/CAN (1 kHz)	$\omega_x, \omega_y, \omega_z$ drone body rates
AZ_pantilt	AZ Sensor Jetson FCC		CAN (100 Hz)	Pan-tilt azimuth position
EL_pantilt	EL Sensor Jetson FCC		CAN (100 Hz)	Pan-tilt elevation angle
EL_direction	EL Direction Sensor FCC	Jetson	CAN	Elevation direction
AZ_command	Jetson FCC Gimbal Controller		CAN	Target azimuth for slaving / IMU correction
EL_command	Jetson FCC Gimbal Controller		CAN	Target elevation for slaving / IMU correction
AZ_motor_cmd	Gimbal Controller Servo	AZ	CAN (Motor ctrl)	AZ servo velocity command
EL_motor_cmd	Gimbal Controller Servo	EL	CAN (Motor ctrl)	EL servo velocity command
BORESIGHT_ACHIEVED	Jetson FCC	ISM	Discrete (24V)	Alignment confirmed — IMU active
ARMING_AUTHORIZATION	Jetson FCC	ISM	Discrete (24V)	Arm command from FCC
ARMED_CONFIRMED	ISM	Jetson FCC	Discrete (24V)	ISM confirms armed state
FIRE_AUTHORIZATION	Jetson FCC	ISM	Discrete (24V)	Fire command from FCC
Safety Open cmd	ISM	Safety Open Solenoid	Discrete 24V	Energize = open safety
Safety Close cmd	ISM	Safety Close Solenoid	Discrete 24V	Energize = close safety
Safety Pin cmd	ISM	Safety Pin Solenoid	Discrete 24V	Energize = arm master trigger
Trigger cmd	ISM	Trigger Solenoid	Discrete 24V pulse	Energize = fire weapon
Recoil pulse	Recoil Sensor ISM		Discrete	Fire actuation mechanical verification
Mech movement signal	Movement Sensor	ISM	Discrete / Analog	20 mm cycle verification — HIGH = complete
MECH_CYCLE_CONFIRMED	ISM	Jetson FCC	Discrete (24V)	Cycle OK — next fire authorized
MECH_FAULT	ISM	Jetson FCC	Discrete (24V)	Cycle incomplete

				— next fire blocked
Weapon temp	Temp Sensor ISM	Analog/CAN		Weapon barrel temperature
Laser range	Laser Range Finder Jetson FCC	Serial/Ethernet		Target distance in meters
Joystick 1	GCS Jetson FCC	RF / Ethernet		Camera slew command
Button 1	GCS Jetson FCC	RF / Ethernet		Target lock activation / release
Arm Button (BTN2)	GCS Jetson FCC	RF / Ethernet		Weapon arm command
Trigger Button (BTN6)	GCS Jetson FCC	RF / Ethernet		Fire command
Display data	Jetson FCC GCS	Ethernet / RF		All HUD symbology and video

18. Full System Hardware Architecture

18.1 Architecture Overview

The following is the complete hardware node diagram of the RWS system, showing every physical component, its role, and all interconnections.



- INPUTS (discrete):
- BORESIGHT_ACHIEVED ◀ Jetson A
 - ARMING_AUTHORIZATION ◀ Jetson A
 - FIRE_AUTHORIZATION ◀ Jetson A
 - Recoil Sensor →
 - Mech Movement Sensor (20mm) →
 - Weapon Temp Sensor →
- OUTPUTS (discrete 24V):
- Safety Open Solenoid
 - Safety Close Solenoid
 - Safety Pin Solenoid
 - Trigger Solenoid

Outputs to Jetson A (discrete):

- ARMED_CONFIRMED, MECH_CYCLE_CONFIRMED, MECH_FAULT, SAFE_CONFIRMED

LAYER 3 – OPTICS & ELECTRO-OPTICAL

OPTIC MASTER
(Camera PC)
Connected to
Jetson A via
Ethernet
GigE, H.264/265

EO DAY CAMERA
(Day Payload)
IR THERMAL CAM
(IR Payload)

Full HD 1080p
10x optical zoom
640x512 LWIR
NETD ≤ 50 mK

- EO/IR CAMERA GIMBAL (mounted on RWS Elevation Platform)
- Self-contained stabilized gimbal (own motors + own IMU)
 - Master in the master/slave fire control loop
 - Reports AZ_cam, EL_cam to Jetson A via Ethernet at 100 Hz
 - Receives Joystick 1 slew commands from GCS via Jetson A

LAYER 4 – MOTION CONTROL & SENSOR FEEDBACK

GIMBAL
CONTROLLER
Receives AZ/EL
commands from
Jetson A via CAN

IMU
(6-DOF)
Gyro: $\leq 0.005^\circ/\text{s}/\sqrt{\text{Hz}}$
Accel: $\leq 100\mu\text{g}/\sqrt{\text{Hz}}$
Rate: 1 kHz
Interface: CAN
→ Jetson A for
IMU stab. loop

AZ SERVO
BLDC 24V
AZ axis
CAN bus

EL SERVO
BLDC 24V
EL axis
CAN bus

(BLDC 24V, 1:60 worm gear)
Cont. torque ≥ 0.15 Nm
Peak torque ≥ 0.25 Nm
Encoder: 4096 PPR absolute

AZIMUTH
SENSOR
Absolute

EL ANGLE
SENSOR
Absolute

EL DIRECTION
SENSOR
Incremental

CAN→FCC

CAN→FCC

CAN→FCC

LAYER 5 – WEAPON MECHANICS & SENSORS

SOLENOIDS (all 24V, controlled by ISM Safety MCU only):

Safety Open Sol.

Safety Close Sol

Safety Pin Sol.
(master trigger)

Trigger Solenoid
(fire actuator)

SENSORS (all report to ISM Safety MCU):

RECOIL MOVEMENT SENSOR

– Discrete pulse → ISM – confirms weapon fired mechanically

MECHANICAL MOVEMENT SENSOR (20 mm displacement sensor)

– Inductive / Hall-effect / LVDT
– Measures bolt/carrier 20 mm travel post-fire
– Discrete HIGH ($\geq 20\text{mm}$) = cycle complete → next fire OK
– No signal or $< 20\text{mm}$ = cycle fault → next fire BLOCKED
– Self-test at SYSTEM_INIT before IDLE_SAFE

WEAPON TEMPERATURE SENSOR

– Analog / digital temperature → ISM
– Blocks fire if temp $> T_{\text{cutoff}}$

LASER RANGE FINDER

– Serial / Ethernet → Jetson A
– Provides target distance for ballistic correction

LAYER 6 – GROUND CONTROL STATION (GCS)

Mobilicom 10" Maxi Controller Pro

Panasonic Toughpad FZ-G2 – 1920×1200 – Intel Core i5/i7

Windows 10 Pro – Integrated Mobilicom RF/TX

CONTROLS:

Joystick 1 → Camera Gimbal AZ/EL slew

Joystick 2 → Reserved (manual override / maintenance)

BTN 1 → Target Lock / Unlock

BTN 2 (SG) → Arm Weapon (safeguard)

BTN 3 → Emergency Abort

BTN 4 → Camera mode (EO/IR/PIP)

BTN 5 → Zoom control

BTN 6 (SG) → Fire Trigger (safeguard)

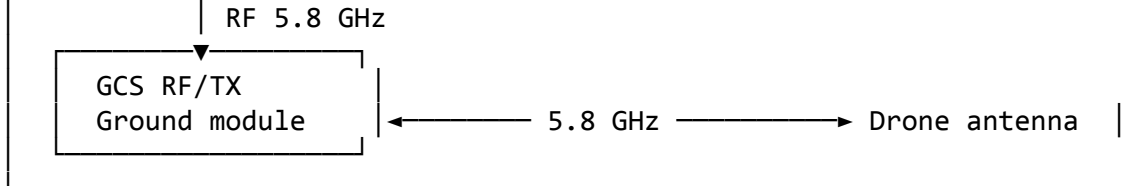
SLI 1 → EO/IR feed toggle

SLI 2 → Zoom level

SLI 3 → IR gain/brightness

TOG 1 → System master power

TOG 2 (SG) → Weapon master enable (physical dual interlock)



18.2 Bus Architecture Summary

Bus	Physical	Devices	Data Rate
Ethernet (GigE)	CAT6 shielded	Jetson A Jetson B Optic Master Laser Range Finder	1 Gbps
CAN-FD	Shielded twisted pair, 120Ω term.	Jetson A Gimbal Controller AZ Servo EL Servo AZ Sensor EL Sensors IMU	5 Mbps
Discrete 24V	Individual signal wires	Jetson A ISM All Solenoids Recoil Sensor Mech Movement Sensor	N/A
RF 5.8 GHz	Antenna (2×SDR)	Drone antenna GCS RF/TX	≥10 Mbps
Video	H.264/H.265 over Ethernet/RF	Optic Master Jetson A GCS	≥8 Mbps per channel
SPI	PCB trace	IMU Jetson A (high-rate IMU data path)	10 Mbps

18.3 Power Distribution

Rail	Source	Consumers	Protection
24V Main Bus	External or battery	All subsystems	Main fuse + reverse polarity
12V Compute	24V DC-DC isolated	Jetson A, Jetson B	Isolated filtered rail
24V Servo	24V direct	AZ servo, EL servo	Per-servo current limit
24V Solenoid	24V direct	All 4 solenoids	ISM-switched, fused
5V Safety	24V LDO independent	ISM Safety MCU only	Independent regulator
12V/5V Optics	24V DC-DC	Optic Master, cameras	Low-noise LDO

18.4 Complete Component List

#	Component	Qty	Interface	Location
1	Jetson A (FCC	1	Ethernet, CAN,	RWS body

	Primary)		Discrete	
2	Jetson B (FCC Redundant)	1	Ethernet	RWS body
3	RF/TX Drone Module	1	RF 5.8 GHz	RWS body
4	Drone Antenna	1	RF 5.8 GHz	RWS top
5	ISM Safety MCU	1	Discrete 24V	RWS body (isolated)
6	Optic Master (Camera PC)	1	Ethernet, MIPI/GigE	EL platform
7	EO Day Camera	1	GigE / MIPI	EL platform
8	IR Thermal Camera	1	GigE / MIPI	EL platform
9	EO/IR Camera Gimbal	1	Ethernet (coords)	EL platform
10	IMU (6-DOF)	1	CAN / SPI	RWS body
11	Gimbal Controller	1	CAN	RWS body
12	AZ Servo (BLDC 24V)	1	CAN	AZ axis
13	EL Servo (BLDC 24V, 1:60 worm)	1	CAN	EL axis
14	Azimuth Sensor (absolute)	1	CAN	AZ axis
15	EL Angle Sensor (absolute)	1	CAN	EL axis
16	EL Direction Sensor (incremental)	1	CAN	EL axis
17	Safety Open Solenoid	1	Discrete 24V	Weapon mount
18	Safety Close Solenoid	1	Discrete 24V	Weapon mount
19	Safety Pin Solenoid	1	Discrete 24V	Weapon mount
20	Trigger Solenoid	1	Discrete 24V	Weapon mount
21	Recoil Movement Sensor	1	Discrete	Weapon mount
22	Mechanical Movement	1	Discrete / Analog	Weapon action

	Sensor (20mm)			
23	Weapon Temperature Sensor	1	Analog / CAN	Weapon barrel
24	Laser Range Finder	1	Serial / Ethernet	EL platform
25	GCS — Mobilicom 10" Maxi Pro	1	RF 5.8 GHz	Operator station
26	GCS RF/TX Ground Module	1	RF 5.8 GHz	Operator station

End of Document — RWS-SW-LOGIC-002 Rev B

This document must be reviewed by the system safety engineer before any implementation begins. All state transition thresholds (0.5° alignment, 500 ms hold, 20 mm travel, 300 ms verification window, T_cutoff values) are subject to revision during system integration testing.